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Advancements in Creole NLP: Overcoming Data Scarcity in Creole Syntactic Parsing

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Motivation and objectives

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1. Part of a 3-year project (Mompelat 2025b) on scalable NLP for Creoles, with real-world applications (e.g., medical translation/interpretation.)
2. Develop robust dependency parsers for ***Martinican Creole*** and ***Haitian Creole***, two under-resourced languages. (Mompelat 2022)
3. Evaluate and compare **data concatenation** and **multi-task learning (MTL)** strategies.
4. Identify pathways to improve parsing accuracy in low-resource settings and guide future annotation work.
5. Ultimately, build better tools for linguists, educators, and Creole-speaking communities. (Mompelat 2025a, Mompelat 2025c)

Martinican and Haitian creoles

- Both are French-lexified.
- SVO Creoles, morphologically reduced: no inflection for gender or number.
- Tense, Mood, Aspect (TMA) expressed with separate morphemes.
- Close syntactic structure; lexical overlap with variation in particles.

The mom liked her sons – La maman aimait ses fils
[MC] Manman-lan **té** enmen **sé** tchanmay-li **a**
[HC] Manman an **te** renmen pitit li **yo**

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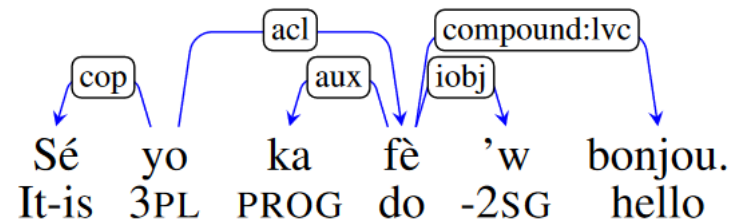


Definition

- What is syntactic Parsing ? Analyzing **syntactic structure** of a sentence.
- Two layers of annotations :
 - **POS tagging** → lexical category of a word
 - Dependency parsing → grammatical role in clause
- Enables: **Machine Translation, Text Prediction, Grammar Tools**, etc.

“Sé yo ka fè’w bonjou” (They are the ones greeting you)

- Treebank:



Treebank efforts in the community :

Universal Dependencies (Nivre et al. 2020) → Standardized annotation framework used in **240+ treebanks** and **100+ languages**.

Token	POS
Sé	AUX
yo	PRON
ka	PART
fè	VERB
,	PUNCT
w	PRON
bonjou	NOUN
.	PUNCT

Definition (cont.)

How Neural Parsing Works:

- Neural models demand large amount of data split into Train, Dev and Test sets.
1. **Training set** → teaches the model (like a class lesson)
 2. **Dev set** → guides learning with corrections (like a practice quiz)
 3. **Test set** → evaluates performance on new data (like a final exam)



Challenge: **Low-resource languages = small datasets**

- How to make the most of this extremely limited corpus?
 - Use related language treebanks (**French**, **HC**, or **MC**)
 - Train models on multiple random splits (seeds)
 - Average performance across seeds to reduce variability

Research Questions

1. Can Haitian Creole improve Martinican Creole parsing?
2. Can Martinican Creole improve Haitian Creole parsing?
3. Does French offer a performance boost for either Creole?
4. Concatenation vs. Multitask Learning: which generalizes better?
5. What are the main parsing challenges that motivate data augmentation strategies for both languages?

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Methodology

Treebanks Used

- French: UD French GSD (Guillaume et al. 2019)
 - High-quality syntactic annotations from a well-resourced language
- Martinican Creole (MC): 236 sentences from news/blogs (2004–2021)
- Haitian Creole (HC): UD Haitian Creole Autogramm (Jagodzińska et al.)

Train	Dev	Test
HC 76	HC 32	MC 36
MC 124	MC 53	MC 59

X 3 seeds

Experimental Designs

- Concatenation → merge HC + MC datasets
- Multi-task Learning (MTL) → train jointly, with task-specific heads

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Methodology – Configuration

Parser: Deep Biaffine Parser via SuPar (Dozat & Manning, 2017)

- Epochs: 70
- Embeddings: TAG (baseline and concatenation) + CHAR + BERT (MTL)

Pre-trained models tested:

- CamemBERT v1 (Martin et al., 2019)
- CamemBERT v2 (Antoun et al., 2024)
- RoBERTa-base (Conneau et al., 2019)
- CreoleVal (Lent et al., 2024) → fine-tuned on Creole NLP tasks

Results - Baseline models

Baseline models trained on a split of Train (45%), Dev (30%), Test (25%)

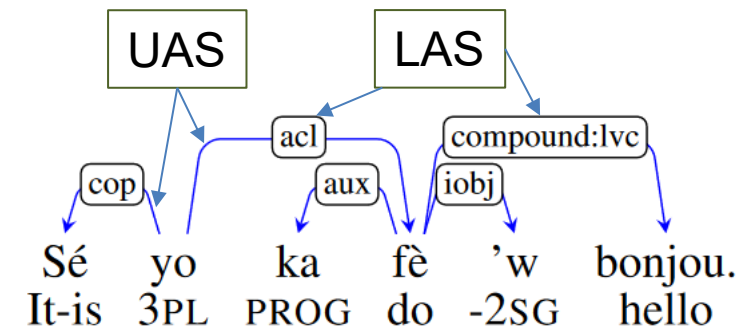
- 3 randomized splits of Train and Dev sets testing on the same test set.
- We tested our 4 pretrained models

Metrics (in %):

- **UAS**: Unlabeled Attachment Score (structure only)
- **LAS**: Labeled Attachment Score (structure + dependency label)

Train	Dev	Test
HC 76	HC 32	MC 36
MC 124	MC 53	MC 59

X 3 seeds



Baseline - MC	UAS	LAS
CamemBERT1	65.63	54.56
CamemBERT2	67.07	56.57
CreoleVal	73.01	64.74
RoBERTa	68.22	58.53

Baseline - HC	UAS	LAS
CamemBERT1	64.89	52.71
CamemBERT2	66.96	56.05
CreoleVal	76.62	68.37
RoBERTa	71.38	58.98

Results – Concatenation Experiments

Setups : **MC+HC**, FR+MC/HC

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Concatenation HC + MC -> MC	Baseline UAS	UAS HC + MC		Baseline LAS	LAS HC + MC
CamemBERT1	65.63	69.64		54.56	58.82
CamemBERT2	67.07	70.19		56.57	60.18
CreoleVal	73.01	76.27		64.74	67.24
RoBERTa	68.22	72.12		58.53	62.28

Concatenation MC + HC -> HC	Baseline UAS	UAS MC + HC		Baseline LAS	LAS MC + HC
CamemBERT1	64.89	72.31		52.71	57.53
CamemBERT2	66.96	74.09		56.05	61.73
CreoleVal	76.62	80.11		68.37	71.9
RoBERTa	71.38	73.98		58.98	62.43

- **All models improved** over baselines
- **Biggest overall gains** from concatenation camemBERT1 and RoBERTa for MC, and CamemBERT for HC

Results – Concatenation Experiments (cont.)

Setups : MC+HC, **FR+MC/HC**

Concatenation FR+MC	UAS Baseline	UAS HC+MC	UAS FR+MC		LAS Baseline	LAS HC+MC	LAS FR+MC
camemBERT1	65.63	69.64	75.46		54.56	58.82	64.16
camemBERT2	67.07	70.19	74.54		56.57	60.18	63.84
CreolEval	73.01	76.27	76.96		64.74	67.24	67.39
roBERTa	68.22	72.12	69.81		58.53	62.28	58.56

- **RoBERTa** underperforms for HC and MC

- French improves **CamemBERT** models for HC and MC

Concatenation FR+HC	UAS Baseline	UAS MC+HC	UAS FR+HC		LAS Baseline	LAS MC+HC	LAS FR+HC
camemBERT1	64.89	72.31	74.87		52.71	57.53	62.58
camemBERT2	66.96	74.09	74.83		56.05	61.73	62.62
CreolEval	76.62	80.11	79.14		68.37	71.9	69.04
roBERTa	71.38	73.98	69.9		58.98	62.43	55.23

- Best UAS/LAS with **Creoleval** for MC but **MC+HC** still outperforms **FR+HC** for HC

- **Creoleval** best overall

Results – MTL Experiments

3 MTL configurations tested:

- **Baseline MTL** = 3 seeds per language
- **HC × MC** = 3 MC + 3 HC tasks jointly trained
- **HC + MC** = 6 mixed-language tasks, test on both MC and HC

MTL experiments MC	UAS Baseline	UAS HC x MC	UAS HC + MC	Best concat model UAS		LAS Baseline	LAS HC x MC	LAS HC+MC	Best concat model LAS
CamemBERT1	70.93	72.18	73.56	75.46		59.11	59.37	62.02	64.16
CamemBERT2	71.77	72	71.54	74.54		60.73	59.49	60.18	63.84
CreoleVal	73.58	75.8	75.49	76.96		60.99	63.93	64.76	67.39
RoBERTa	70.01	70.96	71.02	72.12		57.5	57.24	59.8	62.28
MTL experiments HC	UAS Baseline	UAS MC x HC	UAS MC + HC	Best concat model UAS		LAS Baseline	LAS MC x HC	LAS MC+HC	Best concat model LAS
CamemBERT1	72.35	72.9	76.06	74.87		60.69	60.06	63.1	62.58
CamemBERT2	74.65	74.39	74.05	74.83		62.32	61.02	61.21	62.62
CreolEval	79.92	78.1	80.06	80.11		68.45	68.12	70.71	71.9
RoBERTa	72.79	72.49	73.31	73.98		60.28	59.24	62.06	62.43

- Creoleval performs best in almost all settings
- CamemBERT benefits most from cross-lingual multitask training
- RoBERTa shows limited improvement



Error Analysis

- UAS/LAS highlight performance but not **model behavior**.
- Let's look at **where models succeed or struggle**.

Why Error Analysis Matters

- Highlights *what the model still gets wrong*
- Prioritizes *which structures* need more annotated examples
- Informs *smart annotation*, not just more annotation

Guides model fusion strategies:

- Per-token voting → all models vote on each head, majority wins.
- Per-tag delegation → choose the model best at a specific dependency type (e.g., let Creoleval predict *mark*, RoBERTa handle *nmod*).
- Multitask architecture → stack or mix model outputs in training using shared or semi-shared heads.

Analysis – Strengths and weaknesses of language models with HC

HC	Strengths	Weaknesses	Overpredictions	Omissions
CamemBERT1	aux, det, case, and nsubj	advcl, acl, ccomp	nmod, obj	obl:arg, fixed, dislocated
CamemBERT2	mark, cop, obl, acl:relcl	compound:svc, parataxis, dislocated	nsubj, det	goeswith, advcl:cleft
CreoleVal	advmod, nmod, root	ccomp, acl, obl:mod vs obl:arg	nmod, dep	compound:svc, parataxis:insert, advcl:cleft
RoBERTa	obj, xcomp, ccomp, conj	dep, dislocated	xcomp, obl:mod, nmod	advcl:cleft, compound:svc, goeswith

CamemBERT: Stable on nominal and verbal core labels BUT weak on embedded clauses and dislocated elements → good anchor.

Creoleval: Broad structural coverage BUT Fails on rare clause-level distinctions → Combine with camemBERT

Roberta: Best parsing depth and clause complexity—but benefits from noise control (e.g., suppress overactive xcomp, nmod).

Analysis – Strengths and weaknesses of language models with MC

Models	Strengths	Weaknesses	Overpredictions	Omissions
CamemBERT1	nsubj, obj, det, punct, root	xcomp, advcl	nmod, obl	compound:lvc, discourse, csubj
CamemBERT2	ccomp, acl	nsubj, amod, det, dislocated		dislocated, nsubj:pass
CreoleVal	obl, dep, advmod	case, mark, compound, flat:foreign	dep	compound, flat:foreign
RoBERTa	xcomp, conj	appos, nmod:poss, advcl, discourse	obl:mod, compound	expl, list, discourse

CamemBERT: Good base for argument structure BUT weak on clausal dependencies → combine with Roberta for better clause modeling.

Creoleval: Safest model for modifier and attachment disambiguation BUT sensitive to rare function tags → use in ensemble voting.

Roberta: Best at complex clause structure BUT introduces noise → needs noise control or label confidence filtering.

Conclusion & Next Steps

What we achieved:

- Benchmarked Creole parsing across 4 transformer models
- Compared concatenation vs. multitask learning
- Identified model-specific strengths & weaknesses
- Highlighted critical targets for data augmentation

Key insights:

- Creole-to-Creole transfer can outperform French transfer
- CreoleVal excels with modifiers and general robustness
- Model fusion strategies show promise for low-resource parsing

Next steps:

- Expand treebanks with targeted annotation
- Implement and test fusion-based parsers
- Explore reinforcement learning or LLM-guided generation for data creation





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